Reg. No. | | | | | | | | | | [12CS63/13CS63]

SIXTH SEMESTER B.E/B.TECH EXAMINATION, JUNE–2016

EMBEDDED SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. What is an Embedded system? Explain the purpose of an Embedded System. List its major application areas and give one example for each. **08**

b. Differentiate the following with an example: **06**

- **i.** Microprocessor and Microcontroller
- **ii.** Embedded system and General Purpose Computers

2. a. What is a cache? Explain the following with diagram. **10**

- **i.** Two level cache system
- **ii.** Direct mapped cache
- **iii.** Set associative cache

b. Explain the pipelined execution of a branch in ARM using a pipeline diagram. **04**

3. a. Write the major components of bus protocol. Explain the burst read transaction with a timing diagram. **08**

b. Describe : **06**

- **i.** Timer
- **ii.** Cross compiler
- **iii.** Logic analyzer

4. a. Give examples of interrupts that can arise from the following sources and briefly explain their use: **08**

- **i.** Internal hardware device sources
- **ii.** External hardware device with internal vector address generation
- **iii.** Software error related sources
- **iv.** Software instruction related sources

b. Define Context, Interrupt latency and Interrupt service deadline. **06**

5. a. Explain how threads and processes are used in RTOS. **10**

b. Define the following: **04**

- **i.** Task
- **ii.** Deadlock
- **iii.** Semaphore
- **iv.** Remote Procedure Call (RPC)

6.	**a.**	Explain with neat diagram, the concept of memory mapped object.	**08**
	b.	Explain the following:	**06**
		i. Message passing	
		ii. Remote Procedure Call for IPC	
7.	**a.**	Explain with a neat diagram, the structure of an IP packet.	**08**
	b.	List and explain the advantages and limitations of Simulator Based Debugging.	**06**
8.	**a.**	Describe in brief, the following tools used in the development of Embedded System.	**08**
		i. Source code engineering tool	
		ii. Integrated Development Environment tool	
		iii. Simulator	
		iv. In-circuit Emulator	
	b.	Explain the working of a device programmer with a diagram.	**06**